

Can Cross-Reality Help Nonspeaking Autistic People Transition to AR Typing?

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Figure 1: Screenshot showing the three supported letterboards and their virtual versions

ABSTRACT

About 30% of autistic people are nonspeakers; they cannot use speech to communicate effectively. Pointing to letters on a letterboard held by a Communication and Regulation Partner (CRP) is one alternative method of expressive communication that some members of this population use. In the training of this method, a CRP delivers engaging and customized lessons. Additionally, the CRP provides regulatory, sensory, and attentional support and also works to strengthen the subject's pointing skills. The goal of this training is to equip individuals with the required skills to be able to type independently. Recent studies have proposed using AR to provide opportunities for nonspeakers to practice the motor skills involved in typing. To use such systems, however, there needs to be

a transition phase where a CRP teaches their subject how to interact with a virtual letterboard. In this paper, we explore the feasibility of using cross-reality, in which a CRP and nonspeaker can interact with the same virtual objects simultaneously, as a possible means of fostering this transition. We report a study involving 5 nonspeaking autistic subjects with diverse motor skills interacting using a virtual HoloLens 2 letterboard system we developed called HoloBoard. All subjects succeeded in pointing to letters correctly or spelling on the virtual board. We report process and design recommendations based on feedback obtained from subjects and their CRPs.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI); Interactive systems and tools;

KEYWORDS

Nonspeaking autistic people, augmented reality, cross-reality, assistive technology

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1 INTRODUCTION

Approximately 30% of autistic individuals are nonspeaking, meaning that they lack the ability to communicate effectively using speech [10]. Many nonspeakers have learned to communicate by spelling via pointing to letters on a letterboard [19]. Teaching this method typically involves a trained Communication and Regulation Partner (CRP), who delivers lessons catering to the interests of a nonspeaker [15]. The CRP holds a letterboard in the field of view of the nonspeaker, as shown in Figure 2. The nonspeaker answers spelling, comprehension, and open-ended questions related to the lesson by pointing at letters on the letterboard, while the CRP reads aloud each letter. It is essential to note that the CRP's role is manifold. First, they monitor the nonspeaker's attention to ensure they are focused on the task at hand. Second, they help mitigate nonspeaker anxiety and frustration [15]. Third, since many nonspeakers move constantly as a sensory regulation mechanism [14], the CRP re-positions the letterboard as needed so that it always remains in the subject's field of view. Finally, the CRP structures letterboard interactions to gradually strengthen the ability of a nonspeaker to isolate a letter and point at it. For example, they may choose to use partial letterboards each having only a subset of letters initially before moving on to a full letterboard.

Many nonspeaking individuals have experienced an improved quality of life as a result of learning to communicate using a letterboard. For example, they have gone on to graduate from college [8, 9, 21] and write best-selling memoirs [14]. The goal for individuals who communicate with support is to develop the skills to be able to type independently, without a CRP holding the letterboard. However, reaching that point requires a protracted process that involves the CRP delivering regular, individualized, and systematic opportunities to practice the required skills (e.g., coordinating gaze and pointing to letters) in contexts that involve meaningful and age-appropriate educational content [20]. Furthermore, not everyone can afford to engage the services of a CRP for long periods. These issues motivate the need for technology-based alternatives.

Recent studies [17] [2] have argued that Augmented Reality (AR) could be an effective means to offer communication and educational opportunities for this population. Specifically, these studies have devised AR applications to complement the usual training sessions with CRPs. The AR application delivers lessons virtually and users can interact with the lessons using virtual buttons. However, for an AR application to be successful for this population, a nonspeaker will need support from a CRP to transition from a physical letterboard to a virtual one. This is especially important given technological limitations of AR systems, such as the absence of tactile feedback and issues with gesture detection and field of view. Furthermore, the use of a letterboard is viewed as a cooperative task between the nonspeaker and their CRP. Therefore,

a CRP's involvement is crucial for nonspeakers to embrace such applications.

We present the first attempt at creating an AR environment, called HoloBoard, that allows a nonspeaker to be trained by a CRP to use a virtual letterboard. HoloBoard utilizes cross-reality (CR) whereby both the CRP and the nonspeaker share a common virtual space to practice interacting with a virtual letterboard. Our current implementation supports up to three devices (HoloLens 2 or Quest Pro), where virtual letterboards, shown in Figure 1, are spatially synchronized in real-time among all devices. This approach allows CRPs to train and observe students as they learn to use the virtual letterboard. To ease transition to AR, our system supports three different virtual letterboards that are exact replicas of what some nonspeakers use during conventional training. The HoloBoard system could provide a digital alternative to traditional letterboard based communication.

To evaluate HoloBoard, we conduct a design critique study with a sample of 5 nonspeakers. Subjects used the HoloBoard to practise spelling single letters and short words in a single brief session. We collect feedback from subjects, their CRP, and a caregiver (family member). We report our findings from the session and the post-session interviews. Overall, all subjects successfully spelled single letters on the HoloBoard. Most subjects spelled complete words as well. Our early findings suggest that with additional systematic practice, nonspeaking autistic individuals can transition to using the HoloBoard, which we expect will be a useful step towards independent typing.

2 RELATED WORK

Augmentative and alternative communication systems provide individuals who cannot communicate using speech with alternative expressive language options [5, 23]. Some commonly used systems for nonspeaking autistic children and adults use iPhone and iPad apps that offer users a finite number of pictures and words to select from [18, 22]. Although these systems provide a means of communicating, they are limited in that they include vocabulary that someone else has chosen for the user [13, 23].

Although the use of immersive technologies applications for autistic users is rapidly emerging, to present, such research leaves out the nonspeaking population. In fact, a 2020 review of AR and autism [4] cites 20 research studies, none of which focus on nonspeaking autistic users. Furthermore, only two studies that utilize wearable AR devices include any nonspeaking autistic participants [16]. It is fair to say that the literature shows limited research in the field of immersive technologies targeting this population. For instance, a recent paper proposed an AR application that provides tailored lessons for nonspeaking autistic students where the interaction with the application is conducted using only holographic buttons [17]. However, this application was limited to a single-user experience which does not support the nonspeaker working with a CRP. To overcome this issue, the researchers proposed a CR extension [2] that allows a nonspeaking user to interact with lessons while receiving support from their CRP. However, this application still uses a holographic button interface, requiring both the nonspeaker and CRP to adapt to a new system. While both

of the aforementioned studies presented proof of concept applications, they also did not report any form of user evaluation. Our work differs in that we do not require users to learn new responses. Instead, our implementation leverages the skills letterboard users already have and allows them to work with a CRP as they would in a regular training session. Furthermore, we also present process and design recommendations for AR-based letterboard applications based on insights from our nonspeaking subjects, their CRPs, and their family members.

3 HOLOBOARD

3.1 Design approach

We followed a design critique approach that actively involved nonspeakers, CRPs, and caregivers. Our process follows three phases: preparation, running sessions, and post-processing [1]. The design critique method has been found to be suitable for HCI studies targeting neurodivergent end-users including autistic people [12], and people with dementia [3], as it allows researchers to collect requirements, observations and feedback from different perspectives [6].

Our process started by recording observations from spelling and training sessions with letterboard users. After multiple ideation and brainstorming sessions among our team, we arrived at the current HoloBoard concept. We pitched this to our local nonspeaking community and their CRPs and received a positive response. Prior to design and implementation, we invited members of this population to try the HoloLens 2 and interact with simple holographic user interface elements to explore the feasibility of near-hand interactions. Most nonspeaking subjects tolerated the device and successfully interacted with the virtual objects, thereby encouraging us to move forward with the HoloBoard concept.

The core of our design philosophy is to facilitate the transition to a virtual letterboard by building on existing strengths of users. Thus, we use three of the most common letterboard models used in North America to create virtual 3D replicas of these boards. Each board has different configurations; a partial board with letters from 'A' to 'I', a second partial board from 'J' to 'R'. (There is also a physical partial board with the letters 'S' to 'Z' but this was not used in the current work.) The third board we used contained all the letters from 'A' to 'Z'. Although the three boards have similar dimensions, the partial boards have larger letters given that there are fewer of them. These partial boards are used in conventional letterboard sessions before moving to a full letterboard. We mimic this same approach in our sessions. To further emulate real world interactions, the HoloBoard can be moved using "grab" and "release" gestures by the CRP.

Based on feedback from occupational therapists who work with nonspeakers, we added sensory substitution features to make up for the lack of tactile feedback in our AR environment. First, when a subject hovers over a letter, the colour of the letter would change and the letter would pop out by 1 cm. If the subject presses the letter, the colour changes again and a clicking sound is produced.

3.2 Cross-Reality

CR technology allows users to interact with virtual objects at different levels of immersion, e.g., head-mounted AR/VR devices or

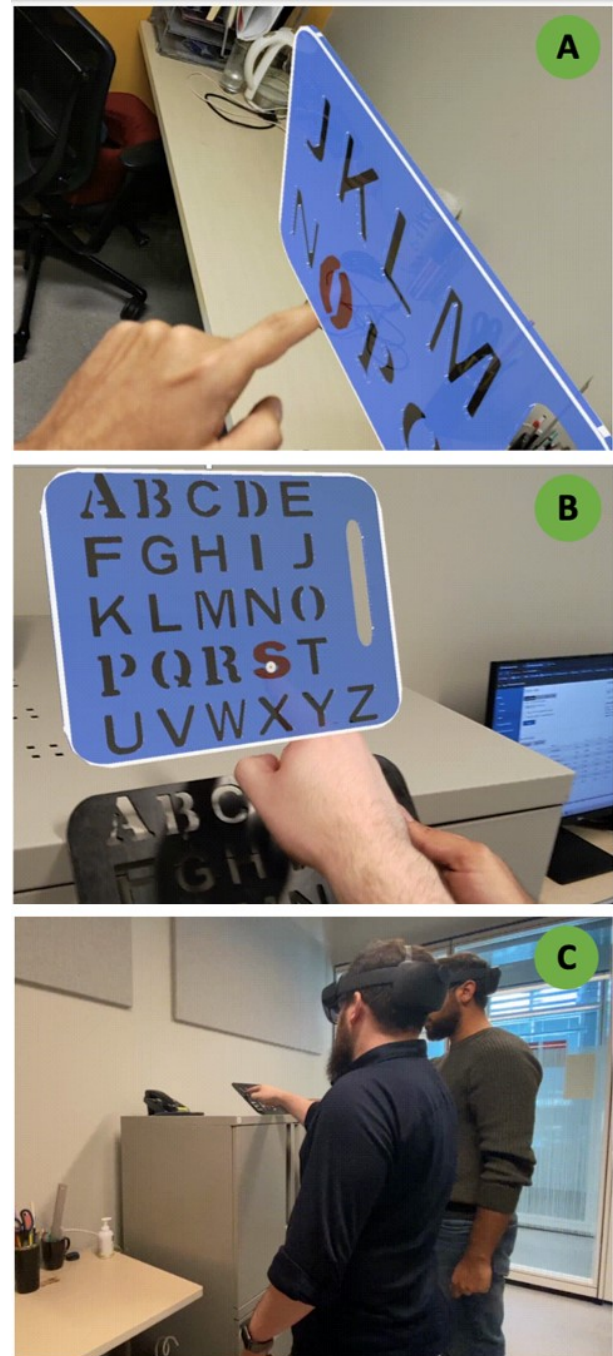


Figure 2: A) a user pointing at a partial virtual board as seen by the CRP. B) Mirroring practice on a full virtual letterboard as seen by the subject. C) Mirroring practice as seen by an external viewer.

simply tablets. Additionally, it allows a virtual workspace to be shared among multiple subjects, each potentially using different types of devices. In our system, we currently support up to three categories of users: nonspeakers, CRPs, and researchers. Figure 2 shows the different views. A CRP can provide verbal prompts to help students focus on tasks. The shared experience also allows them to model virtual letterboard interactions for their subject. A researcher can join the shared session if needed to assist the CRP with technical aspects, e.g., help with moving the virtual letterboard. We have tested the HoloBoard on the HoloLens 2, the Meta Quest Pro (using the "pass through" feature), and the iPad.

3.3 Shared workspace

One of the main challenges we faced was to align the virtual letterboard across all devices in real time and with a high degree of accuracy. To achieve this, we created a novel networking solution called SimpleShareXR, which is designed to be simple to setup and use during our human subject sessions. Specifically, our implementation uses the Unity game engine and the Photon Unity Networking framework for deployment on the HoloLens 2. To create a shared coordinate system among all the users in the shared workspace, we use the Microsoft Spatial Anchors cloud service. This allows us to establish a set of shared control points. Then, SimpleShareXR uses these control points for calculating a shared coordinate system for each user. Finally, it converts the movements of game objects in each user's unique coordinate system to the calculated shared coordinate system. We have tested SimpleShareXR extensively and have open-sourced it for others [11].

4 EVALUATION SESSIONS

Prior to user evaluation, we tested our prototype in-house with actor subjects in order to identify design and implementation issues. We found it useful to add the researcher as a third user to the cross-reality environment who could perform setup and customization tasks in order to reduce the cognitive load on the CRP (many of whom have never used AR before). Furthermore, the researcher can video record their perspective showing the user and the CRP both in the same frame which can provide important insights about the session.

The study was approved by the ethics board of our university. To test the HoloBoard with human subjects, we recruited 5 male nonspeakers. The ages of subjects 5 to 1 were 9, 9, 17, 18, and 24 at the time of the study. The number of years these subjects had been learning to use a physical letterboard were 1.0, 4.0, 7.0, 1.5, and 2.5, respectively. All subjects had a clinical diagnosis of autism. Questionnaires sent to their parents confirmed that they did not use speech to communicate. S1 had their session at their home with a family member serving as their CRP. The other subjects were seen individually at the office of their CRP. All subjects except S5 had the opportunity to experiment with the HoloLens 2 in a dedicated session that occurred before the HoloBoard test session.

For this study, we used the HoloLens 2 device. We note that Microsoft does not recommend the HoloLens 2 device for those younger than 13gi due to the inaccuracy of eye gaze tracking for this sub-population. Since we did not exploit eye gaze tracking in this work, we decided to relax this limit.

Each subject was asked to complete two different exercises. First, subjects would choose a partial letterboard (A-I or J-R) and practice "mirroring spelling," which required finding the letter on the physical board the CRP was holding and then on the virtual board hovering in their field of vision (Figure 2, middle panel). After some practice, the CRP would remove the physical board and ask the subject to find single letters on the virtual board only. Next, the CRP invited the subject to engage in mirroring from the full physical letterboard to the full virtual board. The final task was to spell complete words on the full virtual letterboard, either with or without mirroring. In some cases, the CRP would ask the subject a question designed to have a single word response, e.g., "Which friend did you meet with yesterday?," and invite the subject to spell the answer on the virtual board. We note that some subjects used only one of the partial boards, preferring to move on quickly to the full board. Furthermore, the CRP occasionally adjusted the position of the virtual letterboard based on observed subject performance. The length of the sessions ranged from 6 to 16 minutes, depending on the number of breaks required and user engagement.

We collected feedback at the end of each session. If they were able and willing, nonspeakers were asked to answer exit questions using the physical letterboard. Responses were collected from the CRPs as well. When time permitted, we invited the family members present to try the prototype themselves and offer us their feedback. First-person recordings from the HoloLens 2 devices were recorded from the nonspeaker, their CRP, and the researcher perspectives. We also recorded an outsider view of the session using an external camera.

During the post-processing phase, we merged the HoloLens 2 recordings obtained from the subject, CRP, researcher, and external observer views. After watching these videos and reflecting on each session, we imported them into a qualitative data analysis tool named NVivo. We followed Braun and Clarke's [7] guidelines to run inductive thematic analysis on the qualitative data. The focus of this phase was to code every user interaction: letter spelling, full word spelling, and any other interactions observed. We grouped these codes into main and sub themes based on the exercise and the type of letterboard used. The coding exercise was carried out by the lead author. In the second phase of the analysis, we focused on investigating three types of events, which we expect will be useful to inform the design of a system for practising the skills needed for communication: successful interactions, unsuccessful interactions, and full-word spellings. During the final stage, we reflect on the feedback collected from all our informants to discern key insights.

5 RESULTS

The video coding resulted in 167 observed interactions in total. Given that session length was not uniform, each subject had different numbers of interactions. To facilitate comparisons, we report our findings in terms of the percentage of total interactions a given participant engaged in rather than the raw number of interactions.

5.1 Partial Boards

All participants used both partial boards except for S2 and S3, who chose to use only one partial board before moving to the full board. As Figure 3 shows, three participants were always successful in



Figure 3: Success rates for all subjects in the single-letter-spelling exercise using the mirroring technique and the virtual board only.

mirroring, i.e., pointing to the letter requested first on the physical board and then on the virtual board. One participant was successful in mirroring 85% of the time, while the other 65% of the time.

When asked to select letters on the partial virtual board without mirroring, S2, S3, and S5 were slightly less successful. The success rate of S3, for example, decreased to 70%. However, S1 and S4 improved their success rate. Note that all subjects scored above 70% in the non-mirrored scenario, which is promising given this was the first time they were attempting to spell on a virtual letterboard.

5.2 Full Board

Figure 4 shows results pertaining to a mirroring exercise on a full virtual letterboard. All subjects tried the mirroring except for S4 who skipped directly to spelling directly on the virtual board (discussed next). From Figure 4, the mirroring exercise on the full board was more difficult for some subjects than on the partial boards. S2 and S3 were successful on all mirroring trials. However, S1 completed only two trials: Their initial attempt to press the letter on the full virtual board was unsuccessful, but their second attempt was correct. This subject then requested a break, after which the CRP moved them directly to spelling full words, as we report in the next section. Finally, S5 was correct only 44% of the time. Interestingly, this subject made mistakes on the physical letterboard as well. On further investigation, we learned that the physical letterboard the subject normally uses is larger than the one we had provided, which led to errors on both the physical and virtual boards. This observation reinforces our design philosophy of emulating the real world environment as closely as possible within AR to promote faster adoption of the technology by nonspeakers.

Full word spelling was the last task given to our subjects and the results are shown in Figure 5. The CRP asked subjects either to spell full words or to answer short questions using only the virtual board. Four participants attempted to do so. S1 and S2 spelled three words each on the virtual board independently, which is to our knowledge the first time a nonspeaking autistic person has spelled independently on a virtual letterboard.

The other two subjects had more difficulty with full word spelling on only the virtual board, but they were successful with varying

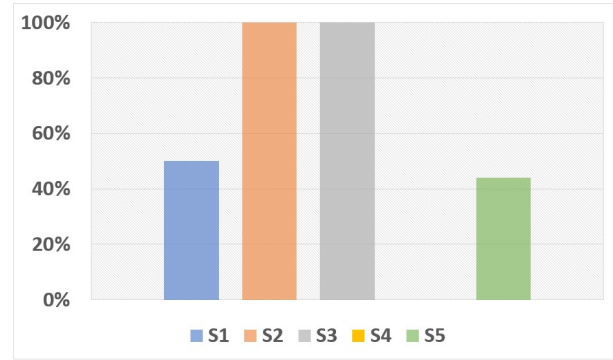


Figure 4: Bar chart showing the success rate for all subjects when practising mirroring spelling on a full letterboard.

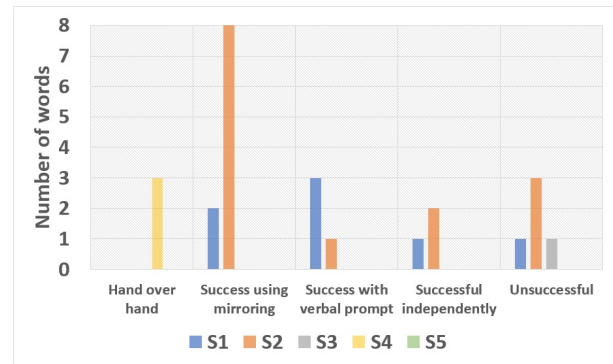


Figure 5: Bar chart presenting results for the full word spelling attempts for all subjects.

degrees of CRP support. When the CRP invited S2 to spell a word on the full physical board and then on the virtual board (similar to the mirroring exercise, but with whole words), they spelled 8 words successfully: 4 words provided by the CRP, e.g., "HoloLens," and "excited," and 4 words that were responses to questions the CRP did not know the answer to, e.g., "What did you do earlier today?". S1 spelled 3 words with CRP support—specifically using mirroring and hand-over-hand modeling from the CRP, i.e., the CRP holding the subject's hand to model an interaction. Overall, these results suggest that with additional practice, these subjects may also be able to spell independently on the virtual letterboard.

5.3 Post-Session Feedback

We conducted semi-structured short interviews, asking subjects for their impressions about the HoloBoard system. We asked follow-up questions when needed. Both the caregiver and CRP were invited to participate as well. Feedback from autistic participants was collected using physical letterboards with the assistance of the CRP. In this section we summarize important findings from these interviews.

Overall, all subjects were satisfied with the experience and they expressed their excitement to participate in future studies. All subjects found the audio feedback (clicking sound) more valuable than the visual changes in the display of the virtual board. Despite the

bright blue color of the virtual boards, some subjects did not find the colour bright enough to stand out from the surrounding environment, which is a critical issue for this population. This feedback calls for further customization of user interface elements to accommodate different user needs. Subjects wondered if the HoloBoard could have a feature that reads out letters as they are pressed and words as they are completed. They pointed out that this feature could provide a greater sense of independence. All subjects at some point during their session tried to move the virtual board themselves (in the current implementation, only the CRP can move it). One subject requested in future iterations that they (as well as CRPs) be allowed to do so. Finally, subjects found the use of the partial virtual boards to be helpful before moving to a full board.

Finding suitable alternatives to compensate the lack of tactile feedback was the main concern of the CRPs and caregivers. Some subjects preferred "swiping" at letters rather than pointing, which meant that sometimes they pointed through the virtual board. One solution recommended by a CRP was to provide an audio-visual feedback for the subject when this happened as a cue to withdraw their hand. Although our subjects favoured the audio feedback, a CRP recommended providing such feedback features as an option that is controlled from the CRP's end to accommodate users who might find the sound overwhelming. Similarly, the CRP wanted the ability to control whether or not a subject can move the virtual board, since some users would find this capability distracting. These comments motivate the need for investigating the user experience requirements for the CRPs themselves.

6 DISCUSSIONS AND DESIGN RECOMMENDATIONS

In most unsuccessful interactions, subjects were pointing close to the correct letter. This is a good indicator of their potential for successfully transitioning to holographic typing. We identified four common error patterns. First, subjects would occasionally point through the virtual board, making it difficult for the subject, the system, and the CRP to detect the exact location of the subject's finger. As noted above, one possible solution for this problem is to implement audio-visual feedback to alert the user of the overshooting. Second, some subjects use a radial pointing technique where they approach the virtual board following a circular trajectory. In this case, the system was not able to detect the subject's input. The CRPs noticed this during the session, and adjusted the virtual board position, which increased the accuracy immediately. The final error pattern is related to the Mixed Reality Tool Kit (MRTK) environment we use, which is designed to track the index finger only. Some users had difficulty keeping the index finger straight. Fortunately, MRTK is capable of tracking all wrist and finger joints. Additional scripts can be implemented to modify the MRTK event registration from the index finger's tip tracking to other finger joints. Finally, some errors were the result of fast subject movements toward the virtual board. By default, MRTK expects a slow approach speed for successful event registration. Overcoming this problem is likely to require more practice from the subject's end as modifying the default value for the pressing speed in MRTK could increase the chances for false positives.

In our case, we only used near-hand interaction limited to touching and pressing. In our analysis we did not separate successful interactions based on whether they were presses or touches. This is because when spelling on a physical board, we observed that pointing and swiping are both common. In general, nonspeakers used the same technique on the virtual board as on the physical board. Thus, it will be important for a system to account for these different spelling techniques. One challenge is that pressing a letter requires precision: the location, the finger gesture, the approach angle, and the approach speed play important roles in registering events successfully. Swiping is similar to "touching" the letter on the virtual board, but registering touches could lead to more false positive. Therefore, it is important to acknowledge the benefits and drawbacks of each of these techniques and take counter measures to reduce the likelihood of false positives.

One of the major advantages of integrating CR into this application is the opportunity for the CRP and nonspeaker to exist in the virtual space together. This allows the CRP to provide effective interventions and prompts during training. In our sessions, this feature became important when students were asked to spell full words. We recorded three types of interventions: mirroring, verbal prompts, and hand-over-hand modeling. When a user had difficulty spelling a word on the complete virtual board, the CRP brought the physical board and suggested mirroring. This intervention helped subjects to successfully spell 12 words in total. Additionally, verbal prompts were helpful to subjects, including encouragements such as "keep going", "what is the next letter", and "finish the word". These type of prompts could be simulated in the application. This may be especially effective when a familiar voice is used, which was suggested by family members and a CRP. Finally, S5 required hand-over-hand modeling with the full virtual board (the CRP guides their hand towards the virtual board and back - slowly and repeatedly - in order to establish muscle memory). Their caregiver reported that the lack of tactile support may have made it difficult for them to recognize where their hand was in space. The hand-over-hand modeling improved the subject's overall success rate. Overall, we found our choice to include the CRP via CR helped subjects get familiar with a new and unfamiliar technology such as the HoloLens 2.

7 CONCLUSIONS AND FUTURE WORK

In this paper, we proposed the first CR-based holographic letterboard spelling system. We conducted a design critique study with a sample of 5 nonspeaking autistic subjects, their CRPs, and caregivers. Our study reveals several novel insights pertaining to the process of training this population on the use of virtual letterboard systems and the design of such systems. Results show that all subjects were highly successful in spelling single letters using only the virtual board. In addition, two subjects spelled full words independently using the virtual board only. The others benefited from verbal prompts, mirroring techniques, and hand-over-hand modeling. These early findings suggest that with systematic training, some nonspeakers may be able to transition to holographic typing. This transition may facilitate effective and more independent communication and, thus, create new social and educational opportunities for this population.

We are currently refining HoloBoard to incorporate the insights gained from this study. Specifically, we will explore ways to overcome technical limitations of the Holo Lens 2 that caused unsuccessful interactions. We will also be carrying out long-term subject training on the HoloBoard system over the course of several months to further characterize the effectiveness of this approach. A key goal of this longitudinal study will be to observe the experience of nonspeakers and track their progress with the virtual board. Furthermore, our future goal is to accommodate remote nonspeakers joining a training session via different immersive technologies. This will allow nonspeakers in locations that do not have CRPs to access letterboard-based lessons and even to interact with peers.

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REFERENCES

- [1] Lorans Alabood, Zahra Aminolroaya, Dianna Yim, Omar Addam, and Frank Maurer. 2022. A systematic literature review of the Design Critique method. *Information and Software Technology* (2022), 107081. <https://doi.org/10.1016/j.infsof.2022.107081>
- [2] Lorans Alabood, Evan Krul, Ali Shahidi, Vikram K Jaswal, Diwakar Krishnamurthy, and Mea Wang. 2022. HoloType-CR: Cross Reality Communication Training for Minimally Verbal Autistic Persons. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 187–190.
- [3] Lorans Alabood and Frank Maurer. 2022. An iot-based immersive smart home system for seniors with neurocognitive disorders. In *Proceedings of the 3d International Workshop on Empowering People in Dealing with Internet of Things Ecosystems, co-located with AVI, Rome, Italy*, Vol. 3172. 10–15.
- [4] Carmen Berenguer, Inmaculada Baixauli, Soledad Gómez, María de El Puig Andrés, and Simona De Stasio. 2020. Exploring the Impact of Augmented Reality in Children and Adolescents with Autism Spectrum Disorder: A Systematic Review. *International Journal of Environmental Research and Public Health* 17, 17, Article 6143 (2020).
- [5] David R Beukelman, Pat Mirenda, et al. 1998. *Augmentative and alternative communication*. Paul H. Brookes Baltimore.
- [6] Eli Blevins, Youn-kyung Lim, David Roedel, and Erik Stolterman. 2007. Using design critique as research to link sustainability and interactive technologies. In *International Conference on Online Communities and Social Computing*. Springer, 22–31.
- [7] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [8] Chad Cardin. 2021. A nonspeaking valedictorian with autism gives her college's commencement speech. *Canadian Broadcasting Corporation News* (2021).
- [9] Bill Chappell. 2022. A nonspeaking valedictorian with autism gives her college's commencement speech. *National Public Radio NPR* (2022).
- [10] Charlotte DiStefano, Wendy Shih, Ann Kaiser, Rebecca Landa, and Connie Kasari. 2016. Communication growth in minimally verbal children with ASD: The importance of interaction. *Autism Research* 9, 10 (2016), 1093–1102.
- [11] Travis Dow, Lorans Alabood, and Diwakar Krishnamurthy. 2023. SimpleShare unity package. <https://github.com/Krishnamurthy-AR-Project/SimpleShare>.
- [12] Christopher Frauenberger, Judith Good, Alyssa Alcorn, and Helen Pain. 2013. Conversing through and about technologies: Design critique as an opportunity to engage children with autism and broaden research (er) perspectives. *International Journal of Child-Computer Interaction* 1, 2 (2013), 38–49.
- [13] Jennifer B Ganz, Theresa L Earles-Vollrath, Amy K Heath, Richard I Parker, Mandy J Rispoli, and Jaime B Duran. 2012. A meta-analysis of single case research studies on aided augmentative and alternative communication systems with individuals with autism spectrum disorders. *Journal of autism and developmental disorders* 42, 1 (2012), 60–74.
- [14] Naoki Higashida. 2013. *The reason I jump: The inner voice of a thirteen-year-old boy with autism*. Knopf Canada.
- [15] Vikram K Jaswal, Allison Wayne, and Hudson Golino. 2020. Eye-tracking reveals agency in assisted autistic communication. *Scientific reports* 10, 1 (2020), 1–10.
- [16] Kamran Khowaja, Bilikis Banire, Dena Al-Thani, Mohammed Tahri Sqalli, Aboubakr Aqle, Asadullah Shah, and Siti Salwah Salim. 2020. Augmented Reality for Learning of Children and Adolescents With Autism Spectrum Disorder (ASD): A Systematic Review. *IEEE Access* 8 (2020), 78779–78807.
- [17] Diwakar Krishnamurthy, Vikram Jaswal, Ahmadreza Nazari, Ali Shahidi, Pranav Subbaraman, and Mea Wang. 2022. HoloType: Lived Experience Based Communication Training for Nonspeaking Autistic People. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–6.
- [18] Elizabeth R Lorah, Christine Holyfield, Jessica Miller, Brenna Griffen, and Cody Lindbloom. 2022. A systematic review of research comparing mobile technology speech-generating devices to other AAC modes with individuals with autism spectrum disorder. *Journal of Developmental and Physical Disabilities* 34, 2 (2022), 187–210.
- [19] Edlyn Vallejo Peña. 2019. *Communication Alternatives in Autism: Perspectives on Typing and Spelling Approaches for the Nonspeaking*. McFarland.
- [20] Edlyn Vallejo Peña. 2019. *Leaders Around Me: Autobiographies of Autistics who type, point, & spell to communicate*.
- [21] David James Savarese. 2019. Coming to my senses. *Autism in Adulthood* 1, 2 (2019), 90–92.
- [22] Monica Tentori, Lizbeth Escobedo, and Gabriela Balderas. 2015. A Smart Environment for Children with Autism. *IEEE Pervasive Computing* 14, 2 (2015), 42–50.
- [23] Janna Van Grunsven and Sabine Roeser. 2022. AAC technology, autism, and the empathic turn. *Social Epistemology* 36, 1 (2022), 95–110.